

MAI-CI Laboratory 3: Genetic Algorithms - Evolution Strategies

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The **Rosenbrock function** is a non-convex function, which is commonly used as a performance test problem for optimization algorithms. Shown in figure 1a, it is defined as

$$f(x, y) = (a - x)^2 + b(y - x^2)^2 \quad (1)$$

In our case we set a and b as 1 and 100 respectively. Table 1 shows the top 10 configurations in our system. We have used random search [1] with 1000 iterations and repeating the same configuration 5 times to take the execution that produces the lowest fitness function value.

#Gen	#Pop	CrossoverFrac	SelectionFcn	InitX	InitY	FVal	X	Y
100	850	0.19696	remainder	[-0.58;0.83]	[-0.29,1.19]	4.874663e-09	0.99995	0.99989
700	800	0.49364	tournament	[-0.25;-0.12]	[-1.84,-1.58]	5.323419e-09	1.00003	1.00006
750	500	0.87429	stochunif	[-1.09;-1.08]	[-0.93,-0.90]	2.692743e-08	1.00038	1.00073
600	450	0.35013	tournament	[-0.90;0.61]	[-0.12,-0.08]	3.200293e-08	0.99933	0.99857
400	850	0.12568	stochunif	[-0.26;-0.24]	[-1.32,0.64]	3.270418e-08	0.99983	0.99965
500	900	0.23846	stochunif	[-0.74;-0.45]	[-0.83,-0.20]	4.577727e-08	1.00006	1.00009
200	950	0.12136	roulette	[0.49;1.23]	[1.23,1.26]	5.931253e-08	0.99980	0.99959
100	950	0.14557	remainder	[-1.26;-0.36]	[-1.99,-0.95]	6.281334e-08	0.99976	0.99952
550	750	0.82664	remainder	[-1.41;-0.86]	[-0.33,0.25]	6.955924e-08	0.98466	0.96955
900	800	0.53466	roulette	[-1.06;-1.01]	[-1.51,1.96]	7.348668e-08	1.00009	1.00021

Table 1: Top 10 configurations with lowest Fitness function value

- **Which and where is the global minimum?** The global minimum is the value obtained evaluating the fitness function (equation 1) within the search space. We know that it has a global minimum at $(x, y) = (a, a^2)$, where $f(x, y) = 0$. Hence, the global minimum, corresponds to $(1, 1)$. Observing the top 10 results, we see that (X, Y) values are both almost equal to 1 and $FVal$ is very close to 0.
- **Which is the parameters combination that give you the best result?** The best results are the ones with the lowest $FVal$. The first row in table 1 shows the best configuration and its obtained results. The best result (along with the other 1000 execution results) is marked in figure 1b.

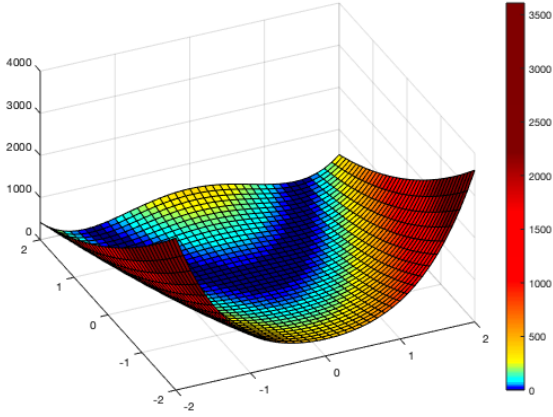
Conclusions

From the results obtained, we can conclude that the global minimum of Rosenbrock function is at point $(1, 1)$. It seems that the number of generations doesn't really affect the result, as the minimum can be reached much faster. However the number of **Population** tends to be large in the top configurations (8/10 are over 750). The **SelectionFcn** neither affects the result, as there is no clear predominant choice among the top results.

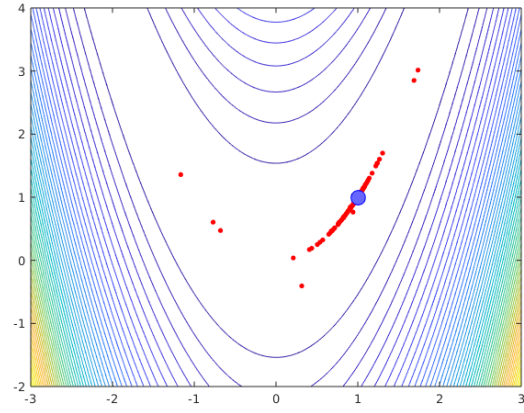
Also, if **PopulationInitRange** doesn't explain a region close to the optimal point, results are farther from the optimal (table 1, penultimate row) because the average distance between individuals is larger and the population diversity in this case is less suited to the problem.

Furthermore, we can observe that some result points lay on a curved valley generated by the fitness function (figure 1b) which make it difficult to converge to the optimal point $(1, 1)$.

This problem is an example of how a simple approach like Genetic Algorithms, with a light representation of the problem and only two operations to perform are able to find solutions for problems of this kind easily and quickly, compared to other approaches such as gradient-based optimizers, which would struggle with it.



(a) 3D Fitness function



(b) Contour plot with GA results of 1000 executions

Figure 1: Fitness Function

References

- [1] James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(Feb):281–305, 2012.